

Student Guide — middle / module-06- moon-phases-tithi

IKS Curriculum

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Student Workbook

Student Workbook — Module 6 Moon Phases & Tithi (Middle)

Name: _____ **Class:** _____ **Section:** _____

This is your workbook for the next two weeks. Write in it freely. Bring it to every class.

Day 1 — The Moon and Why It Changes

The Moon does not change. Our view of it changes.

The eight phases (sketch each):

#	Phase	Sketch
1	new moon	
2	waxing crescent	
3	first quarter	
4	waxing gibbous	
5	full moon	
6	waning gibbous	
7	last quarter	
8	waning crescent	

Synodic month: _____ days (the period from one new moon to the next).

My exit-ticket answer: _____

Moon Diary (run across Days 1–10)

Sketch the Moon's shape each night. Add date/time/cloud cover. From Day 3 onwards, add the tithi.

Night #	Date	Time	Sketch	Pakṣa	Tithi	Cloud
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Day 2 — The Two Pakṣas

A lunar month has two halves:

Sanskrit	English	Moon is
<i>śukla pakṣa</i>	bright fortnight	waxing (growing)
<i>kṛṣṇa pakṣa</i>	dark fortnight	waning (shrinking)

Two anchor days:

- *amāvasyā* — new moon. The Moon and Sun are in the same direction in the sky.
- *pūrṇimā* — full moon. The Moon is opposite the Sun, fully illuminated.

My exit-ticket answer: _____

Day 3 — The 15 Tithis

A tithi is **1/30 of a synodic month** — the time during which the Moon gains 12° of elongation on the Sun.

#	IAST	English
1	<i>pratipadā</i>	first
2	<i>dvitīyā</i>	second
3	<i>ṭṛtīyā</i>	third

#	IAST	English
4	<i>caturthī</i>	fourth
5	<i>pañcamī</i>	fifth
6	<i>ṣaṣṭhī</i>	sixth
7	<i>saptamī</i>	seventh
8	<i>aṣṭamī</i>	eighth
9	<i>navamī</i>	ninth
10	<i>daśamī</i>	tenth
11	<i>ekādaśī</i>	eleventh
12	<i>dvādaśī</i>	twelfth
13	<i>trayodaśī</i>	thirteenth
14	<i>caturdaśī</i>	fourteenth
15	<i>pūrṇimā</i> / <i>amāvasyā</i>	full / new moon

(Stapled formative quiz Part A here)

Day 4 — Build a Tithi Calendar

(See Activity 01 worksheet — paste or staple it here.)

This month: _____

- *Amāvasyā* falls on: _____
 - *Pūrṇimā* falls on: _____
 - Festival(s) I noticed: _____
-

Day 5 — Lunar Months and Festivals

The 12 lunar months:

#	Lunar month	Roughly corresponds to
1	<i>Caitra</i>	March–April
2	<i>Vaiśākha</i>	April–May
3	<i>Jyeṣṭha</i>	May–June
4	<i>Āṣāḍha</i>	June–July
5	<i>Śrāvaṇa</i>	July–August
6	<i>Bhādrapada</i>	August–September
7	<i>Āśvina</i>	September–October
8	<i>Kārtika</i>	October–November

#	Lunar month	Roughly corresponds to
9	<i>Mārgaśīrṣa</i>	November–December
10	<i>Pauṣa</i>	December–January
11	<i>Māgha</i>	January–February
12	<i>Phālguna</i>	February–March

(Stapled formative quiz Part B here)

My family's festivals — month, pakṣa, tithi:

Festival	Month	Pakṣa	Tithi
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My project plan:

- Lunar phenomenon: _____
- Format: _____
- Partner: _____
- Source(s) I'll cite: _____

Day 6 — The Pañcāṅga

#	Limb	What it tracks
1	<i>tithi</i>	Moon's elongation from Sun
2	<i>vāra</i>	day of the week
3	<i>nakṣatra</i>	Moon's position in the star background
4	<i>yoga</i>	sum of Sun + Moon longitudes
5	<i>karaṇa</i>	half-tithi

Today's pañcāṅga snapshot:

- Date: _____
- Tithi: _____
- Vāra: _____
- Nakṣatra: _____
- Yoga: _____
- Karaṇa: _____

Day 7 — Eclipses

Solar eclipse: Sun — Moon — Earth, on _____. Moon's shadow falls on Earth.

Lunar eclipse: Sun — Earth — Moon, on _____. Earth's shadow falls on Moon.

Why not every month? The Moon's orbit is tilted by ~_____° from Earth's orbit.

⚠ SAFETY: NEVER look directly at a solar eclipse. Use ISO 12312-2 certified eclipse glasses or pinhole projection.

(Eclipse-geometry diagram pasted here from Activity 02)

Days 8–9 — Project Work

Today's progress (Day 8):

Today's progress (Day 9):

Day 10 — Reflection

1. What did you set out to learn or make, and what did you actually end up with?

2. What is the most surprising thing you discovered while making this?

3. If you had three more days, what would you do differently?

Glossary (your own, building up)

Sanskrit	Meaning	First met
<i>tithi</i>	lunar day; 1/30 of a synodic month	Day 3
<i>pakṣa</i>	fortnight, half-month	Day 2
<i>śukla pakṣa</i>	bright (waxing) fortnight	Day 2
<i>kṛṣṇa pakṣa</i>	dark (waning) fortnight	Day 2
<i>amāvasyā</i>	new moon	Day 2
<i>pūrṇimā</i>	full moon	Day 2
<i>māsa</i>	lunar month	Day 5
<i>nakṣatra</i>	lunar mansion (1 of 27)	Day 6
<i>vāra</i>	weekday	Day 6
<i>yoga</i>	sum of Sun + Moon longitudes (1 of 27)	Day 6
<i>karaṇa</i>	half-tithi	Day 6
<i>pañcāṅga</i>	five-limbed almanac	Day 6
<i>adhika māsa</i>	intercalary “extra month”	Day 5

Add new terms as you meet them.

Day 1 — The Moon and Why It Changes

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain, with a physical model, why the Moon appears to change shape over a ~29.5-day cycle, and will name the synodic month as the underlying period.

Materials

- A torch / flashlight (the brighter the better — bring spare batteries)
- A small white ball (a table-tennis ball or a styrofoam ball about 4 cm across)
- A darkened classroom (curtains drawn or lights off for ~10 min)
- Whiteboard
- Notebooks

Vocabulary introduced today

- *synodic month* — the ~29.53-day period from one new moon to the next; the basis of the lunar calendar.

(Sanskrit terms start tomorrow. Day 1 is observational and physical.)

Lesson flow

Warm-up (5 min)

- Opening question on board: “*Did you see the Moon last night? What shape was it?*”
- Take 4–5 students’ answers. Sketch each shape they describe on the board (don’t correct yet — just record).
- “*Why is it sometimes a full circle, sometimes a thin sliver, sometimes nothing at all? That’s what this module answers.*”

Core (20 min)

1. **The Moon is a ball lit by the Sun.** Hold up the white ball. “*This is the Moon — a rocky sphere, about a quarter the size of the Earth.*” Shine the torch on it from one side. “*The Sun lights up only one half of the Moon. Always. Half is day, half is night.*”
2. **Why the shape changes — the demo.** Have one student stand still in the centre of the room (= Earth). One student stands ~3 m away holding the torch on (= Sun). You walk slowly around the Earth-student in a wide circle, holding the ball at arm’s length so the Earth-student can see it.

At each of 8 stopping points, ask: “*What fraction of the lit half of the ball can Earth see?*”

- **Position 1 (Moon between Earth and Sun):** Earth sees the dark side. → *new moon*
 - **Position 2 (45° on):** A thin sliver visible on one side. → *waxing crescent*
 - **Position 3 (90°):** Half of the disc visible. → *first quarter*
 - **Position 4 (135°):** Most of the disc visible. → *waxing gibbous*
 - **Position 5 (Earth between Sun and Moon):** Earth sees the fully lit side. → *full moon*
 - **Positions 6–8:** Reverse — gibbous, half, crescent → *waning*
3. **Key insight on the board:**
The Moon doesn’t change. The angle from which we see it changes.
 4. **The cycle takes ~29.5 days.** This is the **synodic month** — the time from one new moon to the next. Write the number 29.53 on the board.
 5. **One question to plant:** “*Earth orbits the Sun in 365 days. The Moon orbits Earth in 29.5 days. Is the lunar ‘month’ the same as a calendar month?*” (No — and that’s why we have leap years for the solar calendar, and a fancier system for the lunar one. We’ll get to that on Day 5.)

Activity (15 min) — Pair sketching

In pairs:

1. One partner is “Sun” (holds an imaginary torch in one direction).
2. The other is “Earth” (stays still).
3. Take a small ball or a balled-up paper and walk it around Earth in 8 stopping positions.
4. At each position, the Earth-partner sketches what shape they see in their notebook.

After 7 minutes, swap roles.

By end of activity, each student has 8 small Moon-shape sketches in their notebook labelled new, waxing crescent, first quarter, waxing gibbous, full, waning gibbous, last quarter, waning crescent.

Wrap-up (5 min)

- **Exit ticket on a slip of paper, 2 min:** “*In one sentence: why does the Moon look different on different nights?*”
- Collect. Read 3 aloud quickly. Correct any “Earth’s shadow makes the phases” answers gently — Earth’s shadow only matters for *eclipses* (Day 7).
- **Tonight’s homework:** Look at the Moon. Note the date, the time, the shape (sketch). Write down the cloud condition. This starts your **10-night Moon diary**.

Student-facing content

The Moon does not change. Our view of it changes.

The Sun lights up half of the Moon. As the Moon orbits Earth, we see different fractions of that lit half. Over ~29.5 days, the visible fraction grows from nothing (new moon), to a half-disc (quarter), to a full disc (full moon), back to a half, and back to nothing.

This 29.5-day cycle is called the **synodic month** — the basis of the lunar calendar that we’ll study in this module.

Homework

- **Tonight (and every night for 10 nights):** Look at the Moon. Sketch its shape in your workbook. Note the date, the time, and “clear / partly cloudy / overcast.” If overcast, look it up on a Moon-phase app or NASA’s published phase calendar.

Differentiation

- **One tier up:** Why is the synodic month (29.53 days) different from the *sidereal* month (27.32 days, the time for the Moon to return to the same star background)? Hint: Earth has moved around the Sun during the lunar month.
- **One tier down:** Skip the 8-position drill. Focus on 4: new, half (waxing), full, half (waning). Build the rest later in the week.

Teacher notes

- Some students will say “the Moon makes its own light.” It does not — it reflects sunlight. Show them this physically by switching the torch off in the demo. The ball goes dark.

- Some will say “the Moon’s far side is permanently dark.” It is not. The Moon’s far side gets just as much sunlight as the near side, on average — we just never see the far side from Earth because the Moon is tidally locked (the same face always faces us).
- For the diary, if students live in a heavily light-polluted city, the thin crescents may be hard to spot in twilight. They can use an app for verification — that’s acceptable.

Citation(s) used in this lesson

- General reference: NCERT Science Class 6, Chapter 17 (“Stars and the Solar System”). No verse citation today.

Day 2 — The Two Pakṣas

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *śukla pakṣa* and *kṛṣṇa pakṣa*, identify the two anchor days *amāvasyā* and *pūrṇimā*, and place yesterday’s observed Moon shape into one of the two pakṣas.

Materials

- Yesterday’s exit-ticket slips (skim before class)
- Board with two columns ready: *śukla pakṣa* and *kṛṣṇa pakṣa*
- The white ball and torch (for a 2-min recall demo)
- Notebooks

Vocabulary introduced today

- *pakṣa* (IAST: pakṣa; lit. “wing, side, fortnight”) — half of a lunar month. There are exactly two pakṣas in every lunar month.
- *śukla pakṣa* (IAST: śukla-pakṣa; lit. “bright fortnight”) — the waxing half: the Moon grows from invisible to full.
- *kṛṣṇa pakṣa* (IAST: kṛṣṇa-pakṣa; lit. “dark fortnight”) — the waning half: the Moon shrinks from full back to invisible.
- *amāvasyā* (IAST: amāvasyā; lit. “dwelling-together”) — new moon; the Moon and Sun are in the same direction in the sky.
- *pūrṇimā* (IAST: pūrṇimā; lit. “fullness”) — full moon; Moon directly opposite the Sun.

Lesson flow

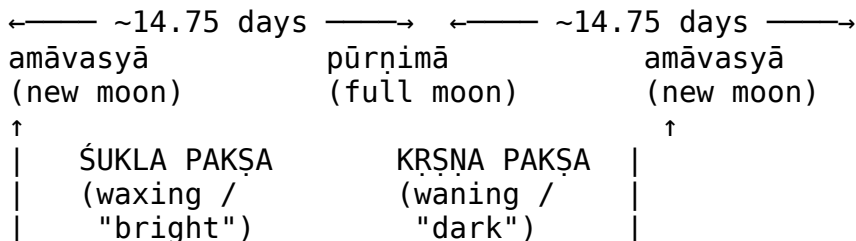
Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What shape?*” — 2 students share.
- Quick recall: ring the question from yesterday’s demo. “*Why does the Moon change shape?*” (Sun lights half; we see different fractions as it orbits.)
- One-sentence reminder: “*The full cycle takes about 29.5 days — the synodic month.*”

Core (20 min)

1. Split the cycle in half.

Draw on the board:



- The fortnight from *amāvasyā* to *pūrṇimā* — when the Moon is **growing** — is the *śukla pakṣa* (“bright fortnight”). The visible lit fraction is increasing each night.
 - The fortnight from *pūrṇimā* to *amāvasyā* — when the Moon is **shrinking** — is the *kṛṣṇa pakṣa* (“dark fortnight”). The visible lit fraction is decreasing each night.
2. **Why these names?** *Śukla* (शुक्ल) means “bright, white”; *kṛṣṇa* (कृष्ण) means “dark, black.” Naming the fortnights by the direction the brightness is heading — toward full or toward new — is precise and useful.
3. **A common confusion to head off:** Neither pakṣa is “good” or “bad.” Both are simply astronomical halves of the lunar cycle. Some festivals fall in *śukla* (e.g. Rāma Navamī, Vaikuṇṭha Ekādaśī), some in *kṛṣṇa* (e.g. Janmāṣṭamī, Mahā-śivarātri).
4. **The two anchor days deserve their names:**
- *amāvasyā* — literally “dwelling-together.” The Sun and Moon are in the same direction in the sky (Moon is between Earth and Sun-ish, near the same celestial longitude). The Moon is up during the day, invisible against the sky’s brightness.
 - *pūrṇimā* — “fullness.” Moon is opposite the Sun. It rises as the Sun sets, stays up all night, sets as the Sun rises. It is fully illuminated.
5. **Connecting to yesterday’s diary.** Hold up your own diary — “*I observed last night, it was a thin crescent right after sunset on the western horizon. That’s a waxing crescent — I’m in śukla pakṣa, just a few days after amāvasyā.*”

Have 3 students share theirs: which pakṣa, how do they know?

Activity (15 min) — Sort the shapes

On the board, draw 8 Moon shapes (you sketched these yesterday in the demo). Number them 1–8 in random order.

In pairs, students:

1. Copy the 8 shapes (small, in a row) in their notebook.
2. Label each as *śukla* or *kṛṣṇa* — and give a reason.

3. Number them in the order they would appear across a lunar month, starting from *amāvasyā*.

Discuss with a different pair (3 min): did you disagree on any? Which?

Wrap-up (5 min)

- **Exit ticket (1 min):** “What is the difference between *śukla pakṣa* and *kṛṣṇa pakṣa*? Use one sentence.”
- Preview Day 3: “Tomorrow we name the 15 days within each *pakṣa* — the *tithis*.”

Student-facing content

A lunar month has two halves — two *pakṣas*.

Sanskrit	English	Direction the Moon is heading
<i>śukla pakṣa</i>	bright fortnight	growing — new moon → full moon
<i>kṛṣṇa pakṣa</i>	dark fortnight	shrinking — full moon → new moon

Two anchor days bookend the *pakṣas*: - *amāvasyā* (new moon) — Moon invisible, “dwelling together with” the Sun - *pūrṇimā* (full moon) — Moon fully lit, opposite the Sun

Neither *pakṣa* is “good” or “bad.” They are astronomical halves of the cycle.

Homework

- Continue the Moon diary. Tonight, add **which *pakṣa* you are in** to your sketch.
- For Day 3, **memorise the names *amāvasyā* and *pūrṇimā* and the two *pakṣa* names.**

Differentiation

- **One tier up:** Look up the meaning of the words *śukla* and *kṛṣṇa* in any Sanskrit dictionary. Are they used only for the Moon, or in other contexts? (They are general-purpose words for “bright/white” and “dark/black.”)
- **One tier down:** Memorise only *amāvasyā* and *pūrṇimā* for now. The *pakṣa* terms can wait one more day.

Teacher notes

- Watch for the misconception that *kṛṣṇa pakṣa* is “the dark side of the Moon” — these are completely different concepts. *Kṛṣṇa pakṣa* is a fortnight of the calendar; the “far side” of the Moon is a separate thing about which face we always see.
- Pronunciation hint: the *ṣ* in *pakṣa* is retroflex — closer to English “puk-sha” than “puk-sa.” Don’t worry about perfection; insist on “pakṣa,” not “paksha-aaa.”

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2 (lunar motion): the lunar month is divided into 30 tithis, distributed 15 each across the two pakṣas. Cited by text name only.

Day 3 — Naming the 15 Tithis

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will define *tithi* in terms of the Moon’s elongation from the Sun (12°), recite the 15 tithi names of a pakṣa in order, and explain why a tithi need not equal a 24-hour calendar day.

Materials

- Board with three columns ready: Number | IAST name | English (ordinal)
- A printed handout of the 15 tithi names (one per student)
- The white ball and torch from Day 1 (one quick re-use)
- Notebooks

Vocabulary introduced today

- tithi* (IAST: tithi; lit. “lunar day”) — $1/30$ of a synodic month; precisely, the time the Moon takes to gain 12° of angular distance (“elongation”) from the Sun.
- elongation* — the angle between the Sun and Moon as seen from Earth (0° at *amāvasyā*; 180° at *pūrṇimā*).

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What pakṣa is it?*” — 2 students share.
- Recall: write on the board — “29.5 days, 2 pakṣas, 2 anchor days.” Ask the class to fill in: how many days in a pakṣa? (~ 14.75 .) How many parts will we divide that into? (15.)

Core (20 min)

1. The tithi as a unit.

“A tithi is one-thirtieth of a full lunar cycle. So a synodic month has 30 tithis. Each pakṣa has 15.”

On the board:

Synodic month (29.53 days)

└ Śukla pakṣa = 15 tithis (1 → 15)
└ Kṛṣṇa pakṣa = 15 tithis (1 → 15)

2. The precise definition (this is important).

“A tithi is NOT one-thirtieth of a day. It is one-thirtieth of the cycle.”

The full cycle covers 360° of elongation (the Moon goes all the way around). Divide 360° by 30 — each tithi is 12° of elongation.

One tithi = the time during which the Moon gains 12° of elongation on the Sun.

On average, this takes about 23 hours and 37 minutes. But the Moon doesn't move uniformly — sometimes faster (perigee, closer to Earth) and sometimes slower (apogee, farther). So a single tithi can last anywhere from ~21.5 hours to ~26 hours.

3. Why a tithi is not a day — three consequences.

- Some tithis “skip” a sunrise. If a tithi starts at 7 a.m. and ends at 5 a.m. the next morning, it never overlaps with a sunrise. This is called *kṣaya* tithi (“decayed” — it never had a sunrise of its own).
- Some tithis “span” two sunrises. If a tithi starts at 5 a.m. on one day and ends at 7 a.m. two days later, it covers two sunrises. This is *vṛddhi* (“growth”).
- Most tithis are tagged to one calendar date — the date during which the tithi is current at sunrise.

4. The 15 names — they are Sanskrit ordinals.

On the board, build the list. Students copy. (Show *śukla* column; same names apply to *kṛṣṇa* with the prefix.)

#	IAST	English (ordinal)
1	<i>pratipadā</i>	“first”
2	<i>dvitīyā</i>	“second”
3	<i>ṭṛtīyā</i>	“third”
4	<i>caturthī</i>	“fourth”
5	<i>pañcamī</i>	“fifth”
6	<i>ṣaṣṭhī</i>	“sixth”
7	<i>saptamī</i>	“seventh”
8	<i>aṣṭamī</i>	“eighth”
9	<i>navamī</i>	“ninth”
10	<i>daśamī</i>	“tenth”
11	<i>ekādaśī</i>	“eleventh”
12	<i>dvādaśī</i>	“twelfth”
13	<i>trayodaśī</i>	“thirteenth”
14	<i>caturdaśī</i>	“fourteenth”
15	<i>pūrṇimā</i> (or <i>amāvasyā</i>)	full / new moon

Note the pattern: 1–10 are simple ordinals; 11–14 are compound (*eka+daśī*, *dvā+daśī*, *trayaḥ+daśī*, *catur+daśī*); the 15th is always the anchor day — *pūrṇimā* at the end of *śukla*, *amāvasyā* at the end of *kṛṣṇa*.

5. **Sanskrit number recall.** *Eka, dvi, tri, catur, pañca, ṣaṣ, sapta, aṣṭa, nava, daśa* — count it together once. The tithi names will stick faster if students see them as ordinals first, Sanskrit terms second.

Activity (15 min) — Tithi drill

Two parts:

Part A (8 min) — paired recall. In pairs, partners alternate: one says a tithi number (1–15), the other names the tithi in IAST. Switch roles every minute. Goal: name all 15 in under 60 seconds by end of activity.

Part B (7 min) — date-to-tithi. Distribute a printed list of 5 dates and the tithi at sunrise on each (lookup done by the teacher from any standard *pañcāṅga* — e.g. for a sample week). Students answer: - *Is this date in śukla or kṛṣṇa pakṣa?* - *What's the Sanskrit name of the tithi?* - *What shape will the Moon be that night?*

Walk around. Check that students are reading the table, not guessing.

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading. Next day's class includes the result.
- **Preview Day 4:** “*Tomorrow we build a tithi calendar for THIS calendar month.*”

Student-facing content

A tithi is the time the Moon takes to gain 12° on the Sun.

- $360^\circ / 30 = 12^\circ$ per tithi.
- Average tithi length: ~23 h 37 min — close to a day but not exactly a day.
- Some tithis skip a sunrise (*kṣaya*); some span two (*vṛddhi*).

Each pakṣa has 15 tithis named as Sanskrit ordinals:

pratipadā, dvitīyā, tṛtīyā, caturthī, pañcamī, ṣaṣṭhī, saptamī, aṣṭamī, navamī, daśamī, ekādaśī, dvādaśī, trayodaśī, caturdaśī, pūrṇimā (or amāvasyā).

Homework

- Memorise the 15 tithi names. Quiz yourself or be quizzed at home.
- Continue the Moon diary — now naming the tithi as well as the pakṣa.

Differentiation

- **One tier up:** Look up the formula for tithi computation: $\text{tithi number} = \text{floor}((\text{Moon_ecliptic_longitude} - \text{Sun_ecliptic_longitude}) \bmod 360) / 12 + 1$. Use an astronomy app (Stellarium is free) to get the longitudes at any moment and compute the tithi yourself. Compare with a published *pañcāṅga*.
- **One tier down:** Memorise only the 6 anchor tithis — *pratipadā, pañcamī, aṣṭamī, ekādaśī, caturdaśī, pūrṇimā/amāvasyā*. Fill the rest in next week.

Teacher notes

- The 12°-per-tithi rule is the single most important computational fact in this module. Drill it.
- Watch for “tithi = day” confusion — every quiz will catch this. The simplest fix: ask “if a tithi started at 7 a.m. and the next started at 5 a.m. the next morning, how many hours was that tithi?” (22.) Repeat with different times until students see that tithi length varies.
- Pronunciation: *ṣaṣṭhī* is hard — “shash-tee.” *Ekādaśī* — “ay-KAH-da-shee.” Don’t rush.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapter 2: the tithi is defined as the period during which the Moon and Sun separate by 12° of arc. Cited by text name only; we do not quote a specific verse.

Day 4 — Build a Tithi Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Each student (or pair) will build a tithi calendar for the current calendar month — labelling each date with its tithi name, pakṣa, and expected Moon shape.

Materials

- One printed page of a current-month *pañcāṅga* per pair (Drik Panchang, Mypanchang, or any standard source). Confirm the source and timezone before printing.
- A blank monthly calendar grid (printed handout — see activity-01)
- Coloured pencils or markers (two colours — one for *śukla*, one for *kṛṣṇa*)
- Notebooks

Vocabulary reused today

- *tithi*, *pakṣa*, *śukla pakṣa*, *kṛṣṇa pakṣa*, *amāvasyā*, *pūrṇimā* (Days 1–3)
- All 15 tithi names (Day 3)

Lesson flow

Warm-up (5 min)

- Daily prompt: “Did you see the Moon last night? What tithi was it?” — 2 students share.
- Hand back yesterday’s formative quizzes briefly (90 sec) — call out the most common miss; correct it once.
- “Today the puzzle is concrete: what’s the tithi for every date this month?”

Core (15 min)

1. **Show one sample row.** Project or photocopy a *pañcāṅga* entry for any one date. Walk through:
 - Date: e.g. “May 22”
 - Pakṣa: e.g. *kṛṣṇa*
 - Tithi at sunrise: e.g. *caturthī*
 - Tithi end-time: e.g. 14:37 (after which *pañcamī* begins)

“So if you went out at 8 p.m. on May 22, the tithi would already be *pañcamī*, not *caturthī*. The tithi listed at the top of a calendar date is the one current at sunrise.”
2. **The pattern of a month — anticipated.** Most months have:
 - A *śukla pratipadā* (the day after *amāvasyā*)
 - A *pūrṇimā* (in the middle)
 - A *kṛṣṇa pratipadā* (the day after *pūrṇimā*)
 - A *kṛṣṇa amāvasyā* (close to the end)
 - 28–31 dated tithis (because of *kṣaya* / *vṛddhi* — see Day 3)
3. **Anatomy of a tithi calendar — the activity sheet.** Walk through the blank grid:
 - One row per date
 - Columns: Date | Day of week | Pakṣa | Tithi name | Tithi end-time | Expected Moon shape
4. **One festival hook.** Find one festival this month from the *pañcāṅga* and announce it: “On *dvitīyā* of *śukla pakṣa* this month, there’s a regional festival called X — let’s see if we agree with the date.”

Activity (20 min) — Build the calendar

This **IS** Activity 01 — see [activities/activity-01-build-tithi-calendar.md](#) for the full set of instructions.

In pairs: 1. Receive one printed *pañcāṅga* page for the current month. 2. Fill in the blank calendar grid for every date of the month. 3. Colour code: *śukla pakṣa* in yellow, *kṛṣṇa pakṣa* in grey/blue. 4. Sketch a small Moon shape for each date (or every 3rd date if running short on time). 5. Circle any *amāvasyā* and *pūrṇimā*.

Walk around. The most common error: students copy the wrong column from the *pañcāṅga*. Have them double-check 2–3 entries with you.

Wrap-up (5 min)

- Pairs hold up their finished calendars briefly. The teacher takes a quick scan.
- **Exit question (1 min, oral):** “Which date this month is *pūrṇimā*?” Class answers in chorus.
- **Preview Day 5:** “Tomorrow we name the 12 lunar months and link them to festivals.”

Student-facing content

Today you built a tithi calendar.

A tithi calendar links each date in a calendar month to: - the *pakṣa* it falls in - the *tithi* current at sunrise - the time the tithi ends - the expected shape of the Moon

Two patterns to notice: - There are usually 1 *amāvasyā* and 1 *pūrṇimā* per calendar month — sometimes 0 or 2 depending on month length. - A few tithis will have unusual entries — a *kṣaya* (skipped) or *vṛddhi* (spanning two days). Note them; they are the system working as designed.

Homework

- Continue the Moon diary. Compare last night's sketch with the calendar you built today — does the shape match the predicted shape for that tithi?
- For Day 5: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Differentiation

- **One tier up:** Look at the *pañcāṅga* for the month BEFORE this one. Find a tithi that was *kṣaya* (skipped). Explain in writing what happened on that date — the previous tithi was running at sunrise, then the next tithi started after sunrise and ended before the next sunrise. So the “skipped” tithi never had its own sunrise.
- **One tier down:** Fill in the date column and the tithi-name column only. Skip the end-times and the Moon shapes for this lesson.

Teacher notes

- **Choose ONE *pañcāṅga* source for the entire class.** Different *pañcāṅgas* may use different sunrise locations or computational systems and give slightly different tithi names for borderline dates. Pick one (Drik Panchang is well-known; Mypanchang.com is also reliable) and stick to it for the whole module.
- Some calendars list two tithis on a single date (e.g. “Caturthī upto 14:37, then Pañcamī”). This is normal. Have students enter the tithi at sunrise, then note the transition.
- The colour-coding (yellow for *śukla*, blue/grey for *kṛṣṇa*) makes the calendar legible at a glance. Insist on it.
- Bring a backup: if printing fails, project the *pañcāṅga* for the whole class to read together. Less efficient but workable.

Citation(s) used in this lesson

- The class *pañcāṅga* source (named on the board): use this consistently. The data is computed using the same astronomical principles described in the *Sūrya-siddhānta* — paraphrased only; no verse cited.

Day 5 — Lunar Months and the Festival Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the 12 lunar months in order, explain how each month is named (after the *nakṣatra* of its *pūrṇimā*), and link at least three festivals to their tithi-and-month positions.

Materials

- Board with a 12-row table ready
- A printed list of major festivals across the 12 lunar months (one per pair)
- Notebooks

Vocabulary introduced today

- *māsa* (IAST: *māsa*; lit. “month”) — a lunar month, ~29.53 days.
- *nakṣatra* (IAST: *nakṣatra*; lit. “lunar mansion”) — one of 27 segments of the ecliptic, each ~13°20’ wide. (Full coverage in a later module — today’s mention is just for naming.)
- *adhika māsa* (IAST: *adhika-māsa*; lit. “extra month”) — an intercalary 13th month inserted approximately every 32.5 months.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- Quick recall using yesterday’s tithi calendar: “*What month did we build the calendar for?*” (May, June, etc.) “*In the Indian lunar calendar, what month is that?*” (Answer: it depends — let’s find out.)

Core (20 min)

1. **The 12 lunar months.** Write the list on the board:

#	Lunar month (IAST)	Roughly corresponds to	Named after
1	<i>Caitra</i>	March–April	nakṣatra <i>Citrā</i>
2	<i>Vaiśākha</i>	April–May	<i>Viśākhā</i>
3	<i>Jyeṣṭha</i>	May–June	<i>Jyeṣṭhā</i>
4	<i>Āṣāḍha</i>	June–July	<i>Pūrva- / Uttara- āṣāḍhā</i>
5	<i>Śrāvaṇa</i>	July–August	<i>Śrāvaṇa</i>
6	<i>Bhādrapada</i>	August– September	<i>Pūrva- / Uttara- bhādrapadā</i>
7	<i>Āśvina</i>	September– October	<i>Aśvinī</i>
8	<i>Kārtika</i>	October– November	<i>Kṛttikā</i>

#	Lunar month (IAST)	Roughly corresponds to	Named after
9	<i>Mārgaśīrṣa</i>	November–December	<i>Mṛgaśīrṣa</i>
10	<i>Pauṣa</i>	December–January	<i>Puṣya</i>
11	<i>Māgha</i>	January–February	<i>Maghā</i>
12	<i>Phālguna</i>	February–March	<i>Phālgunī</i> (Pūrva / Uttara)

2. **Why these names? The naming convention.** A lunar month is named after the *nakṣatra* in which the *pūrṇimā* (full moon) of that month falls. The Moon at full moon is opposite the Sun, so it sits in a specific star background — the *nakṣatra*. The lunar month inherits its name from that *nakṣatra*.

3. **Two month-counting conventions.** This is a real difference and worth noting:

- *Amānta* — the lunar month ends at *amāvasyā*. Common in South and Western India (Maharashtra, Karnataka, Andhra, Tamil Nadu, Gujarat).
- *Pūrṇimānta* — the lunar month ends at *pūrṇimā*. Common in North India (Uttar Pradesh, Bihar, North Indian Hindi belt).

So the same fortnight can be called by different month names under the two conventions! For example, the *kṛṣṇa pakṣa* before Janmāṣṭamī is “*Śrāvaṇa kṛṣṇa pakṣa*” under *pūrṇimānta* and “*Bhādrapada kṛṣṇa pakṣa*” under *amānta*. Both are correct — they are counting from different anchor points.

4. **Linking festivals to tithis — the festival table.** Write 6 examples:

Festival	Lunar month + pakṣa + tithi	What it celebrates / who observes
<i>Ugādi</i> / <i>Gudi</i> / <i>Pādvā</i>	<i>Caitra</i> śukla pratipadā	Lunar new year (in much of South / West India)
<i>Rāma Navamī</i>	<i>Caitra</i> śukla navamī	Birth of Rāma
<i>Akṣaya Tṛtīyā</i>	<i>Vaiśākha</i> śukla tṛtīyā	Auspicious for beginnings
<i>Guru Pūrṇimā</i>	<i>Āṣāḍha</i> pūrṇimā	Honouring teachers
<i>Janmāṣṭamī</i>	<i>Bhādrapada</i> (or <i>Śrāvaṇa</i> , depending) kṛṣṇa aṣṭamī	Birth of Kṛṣṇa
<i>Dīpāvalī</i> / <i>Dīwālī</i>	<i>Kārtika</i> amāvasyā	Festival of lights
<i>Mahā-śivarātri</i>	<i>Phālguna</i> (or <i>Māgha</i>) kṛṣṇa caturdaśī	Great night of Śiva
<i>Holī</i>	<i>Phālguna</i> pūrṇimā	Festival of colours

Every festival here has a precise tithi-and-month coordinate. That's the whole point of the *pañcāṅga* system.

5. **The intercalation puzzle — *adhika māsa*.** 12 lunar months $\times$ 29.53 days = 354.36 days. The solar year is 365.24 days. So the lunar calendar drifts behind by ~11 days a year. After 3 years that's 33 days — about a whole month. To re-align, the Indian system inserts an *adhika māsa* ("extra month") roughly every 32.5 months. This is what makes the Indian calendar **lunisolar** rather than purely lunar. (More on this in the cross-cultural lesson, Day 9.)

Activity (15 min) — Festival mapping + formative quiz

Part A (8 min) — pair work. Each pair gets the printed festival list (about 12 festivals). Their job:

1. Mark which lunar month each festival falls in.
2. Identify the tithi and pakṣa.
3. Find ONE festival that no one in the pair had heard of before. Write its month + pakṣa + tithi.

Part B (7 min) — formative quiz, see quizzes/formative.md Part B. Collect and mark within 24 hours.

Wrap-up (5 min)

- One student names a festival their family celebrates; the class names the tithi+pakṣa+month. Two rounds.
- **Project decision deadline:** by the end of today, every student has a project partner (or has confirmed individual work) and a tentative topic. Sticky note on the board.

Student-facing content

There are 12 lunar months. Each is named after the *nakṣatra* in which its *pūrṇimā* falls.

Caitra, Vaiśākha, Jyeṣṭha, Āṣāḍha, Śrāvaṇa, Bhādrapada, Āśvina, Kārtika, Mārgaśīrṣa, Pauṣa, Māgha, Phālguna.

Two conventions: months end at *amāvasyā* (*amānta*) or at *pūrṇimā* (*pūrṇimānta*). Different regions, different conventions — same astronomy.

Every Indian festival sits at a precise (month + pakṣa + tithi) coordinate. The *pañcāṅga* tells you exactly when.

Homework

- Continue the Moon diary.
- Submit your **project plan paragraph** — 5–8 sentences stating which lunar phenomenon you'll cover (phases / eclipses / intercalary month / something else), in what format, and what your audience walks away with.

Differentiation

- **One tier up:** *Adhika māsa* deep-dive. Look up the rule: an *adhika māsa* is a lunar month in which no *saṁkrānti* (Sun’s entry into a new zodiac sign) occurs. Why does the rule produce ~7 extra months in 19 years?
- **One tier down:** Memorise only the first 6 month names this week. The rest can come during the project.

Teacher notes

- The *amānta* / *pūrṇimānta* distinction confuses everyone the first time. It’s OK to leave students slightly fuzzy on it — the key insight is that conventions vary regionally, and “the calendar” isn’t one fixed thing.
- Some students will know a festival under a regional name (e.g. *Gudi Pādvā* vs. *Ugādi* — both *Caitra śukla pratipadā*). Honour both.
- The “12 months are named after nakṣatras” point doesn’t need full nakṣatra coverage today — that comes in a later module. Today we just say *what* the naming convention is.

Citation(s) used in this lesson

- General reference: *pañcāṅga* sources (named in class). Paraphrase from *Sūrya-siddhānta*, chapter 14 (computation of *adhika māsa*) — cited by text and chapter name only.

Day 6 — The Pañcāṅga: Five Limbs

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will name the five limbs (*aṅga*) of the *pañcāṅga* — tithi, vāra, nakṣatra, yoga, karaṇa — and explain in one sentence each what each limb tracks.

Materials

- The *pañcāṅga* sample page used on Day 4
- Board with a 5-row table ready
- Notebooks

Vocabulary introduced today

- *pañcāṅga* (IAST: pañca-aṅga; lit. “five limbs”) — the traditional Indian almanac listing five astronomical quantities for each day.
- *aṅga* (IAST: aṅga; lit. “limb, part”) — one of the five components.
- *vāra* (IAST: vāra; lit. “day of the week, occasion”) — the weekday.
- *yoga* (IAST: yoga; lit. “joining, combination”) — here, one of 27 astronomical divisions based on the sum of Sun and Moon longitudes. (Distinct from yoga as in Patañjali’s discipline — same word, different application.)

- *karaṇa* (IAST: karaṇa; lit. “doing, half-portion”) — a half-tithi. There are 60 karaṇas in a synodic month.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*So far, we’ve spent five days on ONE limb of the pañcāṅga — the tithi. The pañcāṅga has FOUR more limbs. Today: a tour.*”

Core (25 min)

1. **The five limbs.** Write on the board:

#	Limb	What it tracks	Number of options
1	<i>tithi</i>	Moon’s elongation from Sun	30 per month (covered Days 3–4)
2	<i>vāra</i>	day of the week	7
3	<i>nakṣatra</i>	Moon’s position against the star background	27
4	<i>yoga</i>	sum of Sun-and-Moon ecliptic longitudes	27
5	<i>karaṇa</i>	half-tithi	11 names cycling (because of how the 60 karaṇas of a month are named)

2. **Limb-by-limb walkthrough.** Spend 3 min on each, with concrete reference to the Day 4 *pañcāṅga* page:

Vāra — the weekday (7). Named after the seven classical bodies: Sun (*Ravi*), Moon (*Soma* / *Indu*), Mars (*Maṅgala*), Mercury (*Budha*), Jupiter (*Guru* / *Bṛhaspati*), Venus (*Śukra*), Saturn (*Śani*). English Sunday/Monday/... has the same Sun/Moon/Saturn pattern — same naming logic from the same classical-period astronomy.

Nakṣatra — lunar mansion (27). The Moon takes about 27.3 days to circle the sky and return to the same star background (the *sidereal* month). Indian astronomers divided the Moon’s path into 27 segments, each ~13°20’ wide, named after the star pattern in that segment (Aśvinī, Bharanī, Kṛttikā, Rohiṇī, ... Revatī). The *pañcāṅga* lists which nakṣatra the Moon is in each day.

Yoga — joining (27). Add the Sun’s ecliptic longitude to the Moon’s. Divide by 13°20’. The result is the yoga number (1–27). Each has a name (Viṣkambha, Prīti, Āyusmān, ...). It’s a derived quantity, useful for timing in traditional astrology.

Karaṇa — half-tithi (11 cycling). Each tithi is split into two halves. The first 7 karaṇas of the month are *cara* (“moving”) karaṇas — Bava, Bālava, Kaulava, Taitila, Garaja, Vaṇija, Viṣṭi. They cycle 8 times through the month. The last 4 are *sthira* (“fixed”) — Śakuni, Catuṣpada, Nāga, Kīṁstughna — and appear once each near *amāvasyā*. So $8 \times 7 + 4 = 60$ half-tithis per month.

3. **Why five limbs?** Each limb encodes a different astronomical fact:

- *tithi*: position of the Moon **relative to the Sun**.
- *nakṣatra*: position of the Moon **against the stars**.
- *yoga*: combined Sun + Moon position.
- *vāra*: solar day count.
- *karaṇa*: a finer division for sub-day timing.

Together they pinpoint the astronomical situation on any given day. The *pañcāṅga* is, in effect, a compact astronomical state vector.

4. **Modern parallel.** A modern astronomy almanac (e.g. an ephemeris like NASA's Horizons system) lists planet positions, lunar phase, day of week, eclipse data, etc. The *pañcāṅga* is exactly this kind of document — older format, same purpose. The major difference: the *pañcāṅga* also carries cultural / ritual tags (auspicious times, fast days), which a modern ephemeris does not.

Activity (10 min) — Read the *pañcāṅga*

In pairs, students take yesterday's *pañcāṅga* page and find:

1. Today's date row.
2. The *tithi* (already familiar).
3. The *vāra* (compare with the English weekday — does it match?).
4. The *nakṣatra*. Try to pronounce it.
5. The *yoga* and *karaṇa* names (these are new — just read them, no need to compute).

Each pair writes one sentence: “Today, [date], the Moon is in [nakṣatra] *nakṣatra*, it is [tithi] of the [pakṣa] *pakṣa*, and the *vāra* is [day].”

Wrap-up (5 min)

- Read aloud 2–3 students' sentences. The class reciprocates with the meanings (“Mars day,” “Saturn day,” etc.).
- **Preview Day 7:** “Tomorrow, the big question — why do eclipses happen on *amāvasyā* and *pūrṇimā*, but not every month?”

Student-facing content

A *pañcāṅga* has five limbs. Each tracks a different astronomical fact.

Limb	What it tracks
<i>tithi</i>	Moon's elongation from the Sun (1 of 30)
<i>vāra</i>	day of the week (1 of 7)
<i>nakṣatra</i>	Moon's position against the stars (1 of 27)
<i>yoga</i>	sum of Sun + Moon longitudes (1 of

Limb	What it tracks
	27)
<i>karaṇa</i>	half-tithi (1 of 11 cycling names)

Knowing all five for a given moment is like having a snapshot of where the Sun and Moon are. A modern astronomy almanac does the same job in a different format.

Homework

- Continue the Moon diary.
- For Day 7: read NASA’s published lunar phase page for the current year. Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Differentiation

- **One tier up:** Compute the yoga number for today. You need the Sun’s ecliptic longitude and the Moon’s ecliptic longitude (from any astronomy app). Add them, divide by 13°20’, take the integer part + 1 — that’s the yoga number. Compare with what the *pañcāṅga* says.
- **One tier down:** Memorise three limbs only — *tithi*, *vāra*, *nakṣatra*. The yoga and *karaṇa* terms can stay as “things I’ve heard of” for now.

Teacher notes

- The *yoga* limb is the most confusing — partly because “yoga” also means the spiritual discipline. Note this explicitly. They are different applications of the same Sanskrit root *yuj* (“to join”).
- The *karaṇa* naming is genuinely complicated. Don’t try to teach all 11 names today — just enough that students know there are half-tithis and they have names.
- This is a content-heavy day. If the class is fading, move the activity to a paired worksheet for tomorrow and use today’s time for slow walkthrough.

Citation(s) used in this lesson

- General reference: the class *pañcāṅga* source. The five-limb framework dates to early Indian astronomical literature (e.g. *Pañcasiddhāntikā* of Varāhamihira, 6th c. CE), but the modern *pañcāṅga* form is later. Paraphrased; no specific verse cited.

Day 7 — Eclipses: Why and When

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain (with a diagram) why solar eclipses occur on *amāvasyā* and lunar eclipses on *pūrṇimā*, why eclipses do not happen every month, and what the safe ways to view a solar eclipse are.

Materials

- A small globe (or a large white ball — the Earth)
- A smaller dark ball (a tennis ball, painted dark — the Moon)
- A torch (the Sun)
- Two metres of string (to mark orbits)
- The activity-02 handout for the diagramming exercise
- Notebooks

Vocabulary introduced today

- *eclipse* — when one body's shadow falls on another (lunar eclipse: Earth's shadow on Moon) or one body passes between two others (solar eclipse: Moon between Sun and Earth).
- *ecliptic plane* — the plane of Earth's orbit around the Sun.
- *nodes* — the two points where the Moon's orbit crosses the ecliptic plane.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*You have a Moon diary now. Has any night's Moon been completely covered by Earth's shadow? Probably not. Why not? That's today.*”

Core (25 min)

Safety first (2 min, before anything else). Write on the board in capitals:

NEVER LOOK DIRECTLY AT A SOLAR ECLIPSE. Ordinary sunglasses do not protect your eyes. You need ISO 12312-2 certified eclipse glasses, or use pinhole projection.

Every student copies this into their workbook. We will revisit at end of class.

1. **The basic geometry — three bodies in a line.**

Set up the demo: torch (Sun) on one side, white ball (Earth) in middle, dark ball (Moon) on the other side. Show the two configurations:

- **Solar eclipse:** Sun — Moon — Earth (in that order). Moon casts a shadow on Earth. This can ONLY happen on *amāvasyā* (because that's the only time the Moon is between Sun and Earth).
- **Lunar eclipse:** Sun — Earth — Moon. Earth casts a shadow on Moon. This can ONLY happen on *pūrṇimā* (because that's the only time the Moon is opposite the Sun).

2. **So why doesn't this happen every month?**

“Every month has an amāvasyā and a pūrṇimā. If geometry were the only thing, we’d have a solar eclipse every new moon and a lunar eclipse every full moon. Clearly we don’t.”

The reason: **the Moon’s orbit is tilted by $\sim 5^\circ$ relative to Earth’s orbit around the Sun.** Show this with the string and the two balls — Earth’s orbit is one flat ring; the Moon’s orbit is a smaller ring tilted by 5° .

Most months, when the Moon is at *amāvasyā* (between Earth and Sun), it’s either above or below the Sun-Earth line — its shadow misses Earth. Same for *pūrṇimā* — Earth’s shadow misses the Moon.

An eclipse only happens when the Moon is at *amāvasyā* OR *pūrṇimā* AND it’s near one of the two nodes (the points where the Moon’s tilted orbit crosses the flat ecliptic plane). That’s why eclipses come in pairs (a solar and a lunar within ~ 14 days) and in “eclipse seasons” twice a year.

3. **How many eclipses per year?**

- Solar: 2 to 5 per year, somewhere on Earth.
- Lunar: 0 to 3 per year.
- From any one location, a total solar eclipse is rare — the path of totality is narrow, and most observers see only partial eclipses.

4. **Solar vs. lunar — what they look like.**

- **Solar eclipse:** the Sun briefly turns into a “ring of fire” (annular) or vanishes completely (total). Daylight goes blue-grey. Birds may roost. Only visible from the narrow shadow strip.
- **Lunar eclipse:** the Moon turns dim coppery-red (“blood moon”) because the only sunlight reaching it is sunlight bent through Earth’s atmosphere. Visible from the entire night-side of Earth. Lasts up to ~ 100 minutes.

5. **Indian astronomical sources knew this.** The *Sūrya-siddhānta* (c. 5th–10th c. CE) gives algorithms for eclipse prediction — start time, duration, magnitude. Āryabhaṭa (5th c. CE) had a clear understanding of eclipse geometry (Moon’s shadow, Earth’s shadow, ecliptic plane) and rejected mythological explanations (the “Rāhu swallowing the Sun” story is older and pre-scientific; classical Indian astronomers had a separate, mathematical account).

Activity (10 min) — Diagram the geometry

This is `activities/activity-02-eclipse-geometry.md` in full. Students:

1. Receive the activity sheet (with two unfinished diagrams — Sun in one position, Earth in another, Moon’s orbit drawn around Earth).
2. Place the Moon (a sticker or a drawn circle) at the eclipse positions: between Sun and Earth (solar) and opposite (lunar).
3. Draw the shadow cones.

4. Mark the two nodes on the Moon's orbit and shade the regions where an eclipse is possible.
5. Below the diagram, in 3 sentences, explain: *why does an eclipse not happen every amāvasyā / pūrṇimā?*

Wrap-up (5 min)

- Read the safety rule aloud as a class. Once more.
- Ask: “*What’s the safest way to watch a solar eclipse?*” (Eclipse glasses with ISO certification; or pinhole projection.) “*What about a lunar eclipse?*” (Look directly — no danger.)
- **Preview Day 8:** “*Tomorrow you start your project.*”

Student-facing content

Solar eclipse: Sun, Moon, Earth in line (Moon in middle). Happens on *amāvasyā*. Moon’s shadow falls on Earth.

Lunar eclipse: Sun, Earth, Moon in line (Earth in middle). Happens on *pūrṇimā*. Earth’s shadow falls on Moon.

Why not every month? The Moon’s orbit is tilted $\sim 5^\circ$ from Earth’s orbit around the Sun. So usually the Moon’s shadow misses Earth (at *amāvasyā*) or Earth’s shadow misses the Moon (at *pūrṇimā*). Eclipses happen only when the Moon is *both* at *amāvasyā/pūrṇimā* *and* close to one of the two nodes (where the two orbital planes cross).

⚠️ **NEVER look directly at a solar eclipse.** Use ISO 12312-2 certified eclipse glasses, or project the Sun through a pinhole onto paper. Sunglasses, smoked glass, photographic film do not work. Lunar eclipses are safe to view directly.

Homework

- Continue the Moon diary.
- Look up: what is the date of the next solar eclipse and the next lunar eclipse visible from India? Note in your workbook.

Differentiation

- **One tier up:** Find the dates of the nodes for this year. (The two nodes drift backward through the ecliptic — they complete one full circuit in ~ 18.6 years. This is the *Saros cycle*’s underlying mechanism.) Why does this give us “eclipse seasons” twice a year?
- **One tier down:** Skip the node-drift detail. Focus on: solar eclipse = *amāvasyā* + alignment, lunar eclipse = *pūrṇimā* + alignment. The “tilted orbit” is the key idea; the nodes are the precise version of it.

Teacher notes

- This is the single highest-safety-risk day in the entire IKS curriculum if there’s a solar eclipse during the school term. Even if there isn’t one this term, drill the safety rule. Eclipses come around.

- Some students will know the *Rāhu* story (*Rāhu* the demon swallowing the Sun/Moon during eclipses). Acknowledge it as a pre-scientific narrative shared by many cultures, then return to the geometric explanation. Classical Indian astronomers like Āryabhaṭa explicitly rejected the *Rāhu* explanation in favour of the shadow-geometry account.
- The “5° tilt” number is the key. Repeat it. Without it, the “why not every month” question has no answer.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapters 4–5 (computation of solar and lunar eclipses): the text gives detailed algorithms for predicting eclipse timing using the geometry of the Moon’s nodes. Cited by text and chapter range only.
- Modern reference: NASA’s eclipse pages (search “NASA Eclipse Website” — official, free).

Day 8 — Project Session 1: Plan and First Artefact

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will produce a one-paragraph project plan with element (lunar topic), format, audience, and citation; and complete a first-draft artefact (poster sketch / video storyboard / slide outline).

Materials

- Project supplies for whatever format students chose: poster paper, markers, A4 sheets, laptops, art supplies
- Printed copy of the rubric (`assessments/rubric.md`) — one per pair
- The class *pañcāṅga* (for reference)
- Days 1–7 student notebooks (their own diaries + worksheets)

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- Hand each pair a printed rubric. “*Read through the rubric — 4 minutes. Bring questions to me.*”

Core (5 min)

Take questions on the rubric. The most common questions:

- “*Do we need to quote a verse?*” — No. A paraphrase of the *Sūrya-siddhānta* (e.g. “the *Sūrya-siddhānta* defines a tithi as 12° of elongation”) is enough. A NASA reference is also acceptable. The rubric cares that your source is real and named.
- “*Can we do TWO topics?*” — One main topic. You may compare to a second briefly.

- “What if our diary is missing some nights (clouds)?” — Note “cloudy, predicted [shape]” for those nights. The diary still counts.

Activity (30 min) — Build

Pairs work on their projects. The teacher rotates and checks each pair has:

1. **Chosen topic** — one of:
 - Moon phases (geometry and the 10-night diary visualised)
 - Eclipses (why, when, safety)
 - Tithi calendar (the one built on Day 4 + an explanation)
 - Cross-cultural lunar calendars (preview of Day 9)
 - Lunar months and festivals
 - Adhika māsa (intercalary month)
2. **Chosen format** — one of:
 - Poster (A2 or larger)
 - Slide deck (5–8 slides, projected)
 - 90-second video
 - 1-page printed story / explainer
 - Live demonstration with the torch and ball
 - Infographic / data visualisation
3. **One source named** — *Sūrya-siddhānta* (paraphrased), NCERT Physics XI, NASA’s phase calendar, the class *pañcāṅga*, etc. **Real sources only.**
4. **By end of class:** a one-paragraph plan + a first-draft artefact (a poster sketch, video storyboard, slide outline).

The teacher intervenes if: - A pair is still debating topic at minute 15 — give them 5 minutes, then assign. - A pair’s source is invented or unverifiable — flag it; help them swap. - A pair has split unequally — re-balance the workload before they leave.

Wrap-up (5 min)

- Each pair submits their plan paragraph + first-draft artefact (phone photo accepted).
- **Tomorrow:** finish the build. The cross-cultural calendar lesson runs in parallel — 15 min content, then 30 min build time. Be ready.

Student-facing content

Today is build day.

By the end of this class, your pair (or you, if solo) should have: - A one-paragraph plan in writing. - A first-draft artefact — poster sketch, slide outline, video storyboard. - Your source named, on the artefact.

Tomorrow you refine. Day 10 you present.

Homework

- Continue the Moon diary (you have 2 nights left).
- Finish at home what you couldn't finish in class. Don't *start* something new — refine.

Differentiation

- **One tier up:** Add a second citation. Two sources is a stronger project — for instance, paraphrase the *Sūrya-siddhānta* AND cite NASA's phase calendar.
- **One tier down:** Reduce scope. Pick ONE phenomenon (phases, eclipses, or the calendar) and explain it in one paragraph + one diagram. That's a complete project.

Teacher notes

- Pairs that are still “researching” by minute 20 are stuck. Push them to start sketching/drafting even if incomplete. The artefact concretises the thinking.
- Phone-photo submissions are fine. Photograph the draft poster against a wall; submit by email or via shared drive.
- A common failure mode: pairs over-design the visual and under-write the explanation. Push toward writing the **3-minute script** by end of class.

Citation(s) used in this lesson

- The rubric ([assessments/rubric.md](#)) is the primary reference for what “good” looks like.
- Source-quality reminder: ONLY real sources. No invented citations.

Day 9 — Cross-cultural Lunar Calendars + Project Build

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min (15 min content + 30 min project)

Learning objective

Students will compare the Indian lunisolar calendar with three other lunar/lunisolar calendars (Islamic Hijri, Chinese, Hebrew) — identifying what is shared (lunar month basis) and what differs (intercalation rules).

Materials

- Board with a comparison table ready
- Project supplies (continued from Day 8)
- Notebooks

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.

- “Many cultures, looking at the same Moon, built different calendars. Today: a 15-minute compare-and-contrast, then back to project build.”

Core content (15 min)

1. **The common starting point.** Every culture observing the Moon notices the ~29.5-day cycle. The question every culture answers: *how do we fit 29.5-day lunar months into a 365.24-day solar year?*

The three resolutions:

- **Ignore the solar year.** Use only lunar months. The months drift through the seasons. (Islamic / Hijri)
 - **Re-align periodically with an extra month.** Lunisolar. (Indian, Chinese, Hebrew)
 - **Ignore the lunar month.** Use only the solar year, fix months at 30–31 days. (Modern Gregorian)
2. **The four calendar table.** Build on the board:

Feature	Indian (Hindu)	Islamic (Hijri)	Chinese	Hebrew
Type	Lunisolar	Pure lunar	Lunisolar	Lunisolar
Months/ year	12 (or 13 with <i>adhika māsa</i>)	12	12 (or 13 with leap month)	12 (or 13 with <i>Adar II</i>)
New month begins at	<i>amāvasyā</i> or <i>pūrṇimā</i> (region-dependent)	First sighting of crescent after new moon	<i>amāvasyā</i> (new moon)	<i>amāvasyā</i> (with calculated rules)
Days per year (average)	~365.25	~354	~365.25	~365.25
Intercalati on rule	Month with no <i>saṁkrānti</i> becomes <i>adhika</i> (extra)	None	Month with no <i>zhōngqì</i> (zhongqi solar term)	7 leap months in 19 years (Metonic)
Drift through seasons	Stays aligned	Drifts ~11 days/year backward	Stays aligned	Stays aligned

3. **A specific example.** *Ramaḍān* (Islamic) shifts ~11 days earlier each Gregorian year. *Caitra śukla pratipadā* (Indian new year, *Ugādi/Gudi Pādvā*) stays in March–April. *Pesach* (Hebrew) stays in spring. *Chūn Jié* (Chinese new year) stays in late January / early February. The lunisolar systems all use intercalation to keep their months in-season; the pure lunar system does not.
4. **Why does any of this matter?** Three reasons:

- **Cultural literacy.** Knowing how other calendars work helps in a multi-cultural society.
- **Astronomical insight.** Each calendar reveals a different choice in handling the same underlying mismatch (29.5 vs. 365.24).
- **Historical depth.** All four systems were refined over many centuries. The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation — convergent astronomical reasoning.

Activity (25 min) — Project build

Pairs continue their projects from Day 8. The teacher rotates with two priorities:

1. **Source check.** Every project has a real, named citation by end of class today. No invented references.
2. **3-minute script check.** Every pair has written or rehearsed a 3-minute presentation script. The artefact alone is not the project; the *explanation* is.

Pairs whose project relates to cross-cultural lunar calendars can use the Day 9 content directly. Others are encouraged to add a one-sentence cross-cultural note (“this Indian system, like the Chinese and Hebrew, is lunisolar…”).

Wrap-up (5 min)

- All pairs submit a near-final draft (photo / file).
- One sentence from each pair on the board: **“Our project covers ____ and our source is ____.”**
- **Tomorrow:** Day 10. 3-minute presentations. Summative quiz.

Student-facing content

Four lunar/lunisolar calendars compared:

- **Indian (lunisolar)** — 12 months, *adhika māsa* every ~32 months.
- **Islamic / Hijri (pure lunar)** — 12 months, no intercalation. Drifts ~11 days/year.
- **Chinese (lunisolar)** — 12 months, leap month every ~32 months.
- **Hebrew (lunisolar)** — 7 leap months in 19 years (Metonic cycle).

The Indian, Chinese, and Hebrew systems independently arrived at lunisolar intercalation. Their rules differ; the underlying problem is the same.

Homework

- Continue the Moon diary (one night left).
- Final touches on your project. Bring everything to class tomorrow.

Differentiation

- **One tier up:** Pick TWO calendars and write a paragraph: under what circumstances does each calendar’s design choice make sense? (Pure lunar — when religious months not tied to season matter most; lunisolar — when agriculture matters; pure solar — when civil administration matters.)

- **One tier down:** Compare just the Indian and the Hijri (pure lunar). One difference, one similarity. That's enough.

Teacher notes

- This content is genuinely cross-cultural. Avoid framing as “ours is best.” All four systems work; they encode different priorities.
- The Metonic cycle (19-year Hebrew cycle) and the Indian intercalation rule converge to about the same accuracy. This is mathematically interesting — students at Senior band can explore it.
- Keep this part to 15 minutes. The project is the major work today.

Citation(s) used in this lesson

- General references for each calendar — use any standard encyclopaedic source (e.g. *Encyclopaedia Britannica* entries on the Hijri, Chinese, and Hebrew calendars). Do not invent specific verse numbers from primary sources.

Day 10 — Final Assessment + Showcase

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min (split: 20 min quiz + 25 min showcase)

Learning objective

Students demonstrate the module's full set of objectives through a summative quiz, and present their projects in 3 minutes per pair to the class.

Materials

- Printed summative quiz (`quizzes/summative.md`) — one per student
- All projects ready to present
- A visible timer (phone, stopwatch)
- The rubric (`assessments/rubric.md`) for marking

Lesson flow

Warm-up (3 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share. (Last time for the diary prompt.)
- “*Two things today — quiz first, then the showcase. Pencils ready.*”

Summative quiz (20 min)

- Distribute the quiz.
- 20 minutes, closed book, individual.
- Strict silence after the start signal.
- Collect at end. Mark over the weekend.

Showcase (20 min)

- Pairs present in turn — 3 minutes each + 1 minute Q&A.
- Visible timer running. Strict cutoff at 3:30.
- Teacher marks on the rubric in real time.
- Audience: students take notes — what's one fact they didn't know, from each pair?

(For a class of 30 students = 15 pairs, this won't fit in 20 minutes. **Either split presentations across Day 9 and Day 10, or schedule a separate showcase session.** See the lesson-plan note.)

Wrap-up (2 min)

- *“You’ve each kept a 10-night Moon diary. You’ve built a tithi calendar. You’ve explained eclipses. You’ve presented a project. Look up tonight — you’ll see the Moon a little differently.”*
- Reflection slips submitted with projects.
- Collect Moon diaries.

Student-facing content

Module complete.

Today: 1. Summative quiz — 20 min, individual, closed book. 2. Showcase — 3 minutes per pair + 1-minute Q&A. 3. Reflection slip submitted.

By end of class, you’ve shown what you learned. Take a breath.

Homework

- **None.** Module complete.

Differentiation

- **One tier up:** Volunteer to be the audience-facing rubric scorer (anonymously) — practice the discipline of assessing peers fairly.
- **One tier down:** Students who couldn't finish their project today have until next class to submit. Don't penalise on Day 10 if the artefact is 80% done.

Teacher notes

- Mark the quiz over the weekend; return within 3 school days.
- Project rubric marks should be visible to students within 5 school days.
- Reflection slips are NOT graded but are read — write one-sentence response on each.
- Keep one or two strong projects (with student permission) as exemplars for next year.

Citation(s) used in this lesson

- N/A — this is assessment day.

Writing Prompts

Writing Prompts — Module 6 Moon Phases & Tithi (Middle)

Eight prompts in mixed modes. Assign as in-class quick-writes (10 min), homework (one paragraph), or as journal entries woven across the module. Each prompt names the lesson day where it fits naturally.

Reflection prompts

- 1. (Day 1, 5 min)** What did you think made the Moon change shape, before today? In 3–5 sentences, describe what you used to believe, and what changed your mind in class today. (If you already knew the geometry, write about a younger sibling or friend’s belief instead.)
- 2. (Day 7, 10 min)** Stand outside for two minutes one evening (or open the window). Look up. Then list the Moon’s position, its shape, and the tithi. In 4 sentences, describe whether the actual sky matched the *pañcāṅga* prediction. (If it didn’t, what might explain the difference?)

Explanatory prompts (in your own words)

- 3. (Day 2 or 3, homework, ~150 words)** Explain to a younger sibling (Grade 4) why a tithi is not the same as a day. Don’t use the word “elongation” — find your own way to say it.
- 4. (Day 6, in-class, 10 min)** Imagine you are explaining the *pañcāṅga* to a friend from another country who has never seen one. In 5 sentences, describe what the five limbs are and why having all five gives you a more complete picture than any one limb alone.

Argumentative prompt

- 5. (Day 6 or Day 7, homework, 1 paragraph)** The modern Gregorian calendar ignores the Moon entirely — months are 30 or 31 days regardless of the Moon’s actual phase. The Indian *pañcāṅga* tracks the Moon precisely but needs an *adhika māsa* every ~32 months to stay aligned with the seasons. Pick a side and defend it in 5–7 sentences: which approach is better for *modern civic life*, and which for *cultural/agricultural life*? Acknowledge the other side’s strongest point before disagreeing.

Creative prompt

- 6. (Day 5 or Day 8, can be project — 1 page)** Write a short scene (50–150 words) in which the Sun, Earth, and Moon have a conversation about why an eclipse is happening today. Each character gets at least one line. End with one of them explaining why this doesn’t happen every month.

Cross-disciplinary prompt

7. (Day 6 or homework, ~120 words) The Indian *pañcāṅga* and a modern astronomy ephemeris (e.g. NASA Horizons) do similar work — both list astronomical facts for each day. Pick another field — geography, music, finance, sports — and describe a similar pairing: one traditional system and one modern system that do the same job in different formats. (Example: traditional Indian musicology’s *rāga* system and modern Western music theory both organise pitch — different formats, same domain.)

Stretch prompt

8. (Optional, advanced students, Day 10 onwards, ~250 words) The *Sūrya-siddhānta* (Indian, c. 5th–10th c. CE) and the *Almagest* of Ptolemy (Greek, 2nd c. CE) both describe lunar motion in detail and both use geometric models with epicycles. They were developed independently. In a single page, argue for OR against the claim: “*Two cultures arriving at similar models of the same sky is evidence that the sky’s behaviour, not the cultures’ biases, drives astronomy.*” Use one specific example to support your argument.

Prompt-to-band notes

- All prompts are usable at Middle band as written.
- For **Primary** band, prompts 1, 2, 3, and 6 can be adapted (shorter, drawing-friendly).
- For **Senior** band, prompts 4, 5, 7, and 8 are essay-quality without modification; add a citation requirement.

How to give feedback

- Short prompts (1, 2): respond with one sentence of acknowledgement, not corrections. The point is observation.
- Explanatory (3, 4): respond with “the part that landed” and “the part still unclear.” Don’t correct grammar unless severely impeding meaning.
- Argumentative (5, 8): use the rubric — does the writing acknowledge the other side, give a specific example, and reach a clear position?
- Creative (6): no grading. Read aloud the best ones if you have permission.

Homework

Homework — Module 6 Moon Phases & Tithi (Middle)

Homework tasks per day. None should take more than 25 minutes for a Middle-band student. Skip a day if it conflicts with another subject’s deadline.

The **Moon diary** is a running task: every night from Day 1 to Day 10, sketch the Moon’s shape, note the date/time/cloud cover, and (from Day 3 onwards) name the tithi.

Day 1 → Day 2

Task: Start your **10-night Moon diary**. Tonight: look at the Moon. Sketch its shape in your workbook. Note the date, time, and “clear / partly cloudy / overcast.” If overcast, look up the phase on a Moon-phase app or NASA’s published phase calendar.

Time: 5 min (looking) + 5 min (sketching).

Day 2 → Day 3

Task: Continue the Moon diary. Tonight, add **which pakṣa you are in** to your sketch.

Bonus: Find out from a parent or a calendar — has anyone in your family observed *amāvasyā* or *pūrṇimā* this month for any reason (fast, ritual, prayer)?

Time: 10 min.

Day 3 → Day 4

Task: Memorise the 15 tithi names (śukla pakṣa). Quiz yourself or be quizzed at home. **Be ready to recite all 15 in under 60 seconds at the start of tomorrow’s class.**

Continue the Moon diary — now naming the tithi as well as the pakṣa.

Time: 15 min.

Day 4 → Day 5

Task: Compare last night’s Moon-diary sketch with the tithi calendar you built in class. Does the shape match the predicted shape for the tithi?

Also: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Time: 10 min.

Day 5 → Day 6

Task: Submit your **project plan paragraph** in writing.

Format: one paragraph, 5–8 sentences, stating: which lunar phenomenon, which format (poster / demo / story / video / infographic / slide deck), and what audience will see/hear/feel.

Time: 15 min.

Day 6 → Day 7

Task: Read NASA’s published lunar phase page for the current year (search “NASA Moon phases [year]”). Note any solar or lunar eclipses listed for this year — when, where visible. Bring 1 example.

Continue the Moon diary.

Time: 20 min.

Day 7 → Day 8

Task: Bring ALL project materials to tomorrow’s class. The build session starts immediately. Anything you forget cannot be retrieved during class time.

Bonus: Look up the date of the next solar eclipse and the next lunar eclipse visible from your location. Note in your workbook.

Time: 15 min.

Day 8 → Day 9

Task: Finish at home what you couldn’t finish in class today. Tomorrow includes the cross-cultural lesson + refining, not starting.

Time: variable — but cap at 30 min, then stop and rest. Done-is-better-than-perfect.

Day 9 → Day 10

Task: Final touches. Make sure you have: - [] The artefact (poster / video file / printed story / etc.) - [] Your script or notes - [] Your completed Moon diary (10 nights) - [] A working version of your 3-minute presentation in your head

Time: 15 min.

After Day 10

No homework. Module is complete. Take a breath. (If your next module begins immediately, check that module’s README for prerequisite reading.)

Optional follow-up: Keep the Moon diary going for another 20 days. By the end you'll have observed (or noted from a calendar) a full month of phases — and you'll know exactly which night to look for the next *pūrṇimā*.